

SOLID CONNECT
Product Concept, Business Model and System Architecture
Client Discussion and Project Planning Document

Prepared for Solid Connect Limited

This document consolidates the business, product, functional and technical concepts discussed for the proposed Solid Connect digital marketplace.

August 2026

1. Executive Summary

Solid Connect is envisioned as a trusted digital marketplace connecting individuals, businesses, agencies and organizations with skilled artisans, former professionals and other service providers. The platform will be delivered through mobile applications for Android and iOS, supported by web-based business and administration portals.

The platform should support the complete service lifecycle: provider discovery, service requests, intelligent matching, quotations, communication, job confirmation, completion, payment, dispute handling and reviews.

The proposed architecture is designed with a target of approximately one million registered users while avoiding unnecessary complexity at launch. The recommended foundation is React Native with Expo for mobile, Next.js for web, NestJS and TypeScript for the core backend, and Supabase PostgreSQL with PostGIS for the data and location foundation.

Proposed positioning

Solid Connect connects people and organizations with trusted artisans and experienced professionals based on skill, location, availability, experience and reputation.

2. Product Vision

The long-term vision is for Solid Connect to become a trusted digital infrastructure for finding and engaging skilled people. The platform can begin as an artisan marketplace and progressively expand into a broader professional services and workforce marketplace.

Direction	Description
Artisan marketplace	Connect customers with plumbers, electricians, carpenters, masons, welders, painters, cleaners, repair specialists and other skilled workers.
Professional services marketplace	Connect customers and organizations with experienced professionals for consulting, advisory, mentoring, technical supervision and project work.
Workforce marketplace	Help organizations source skilled people for projects, recurring work and short-term engagements.

3. Business Model

Solid Connect is best positioned as a two-sided service marketplace with a business-to-business segment. The demand side consists of individuals, businesses, agencies and organizations. The supply side consists of artisans, skilled workers and experienced professionals.

3.1 Value Proposition

Stakeholder	Value
Individuals	Convenient and trusted access to service providers.
Artisans	Digital visibility, service requests, professional identity and reputation building.
Former professionals	A channel to monetize experience through consulting, mentoring and project work.
Businesses and agencies	Structured provider sourcing, quotations, projects and workforce management.
Solid Connect	A scalable marketplace with transaction, subscription and premium revenue opportunities.

3.2 Revenue Model

Revenue stream	Concept
Transaction commission	Percentage of completed service transactions or contracts.
Business subscriptions	Paid plans for organizations requiring advanced project and workforce features.
Provider premium services	Optional visibility, promotion, analytics and premium tools.
Verification services	Selected paid verification services where commercially and legally appropriate.
Large project facilitation	Negotiated fees for large workforce or professional engagements.
Recurring service contracts	Commission or service fees from recurring engagements.

More providers

- > more skills available
- > more customers can find providers
- > more jobs
- > more provider earnings
- > more providers join
- > more reviews and marketplace data
- > better matching
- > more customers

4. Target Market and Launch Strategy

A focused launch is recommended rather than attempting to serve every category and location at once. Greater Accra can serve as an initial market, followed by expansion into other major Ghanaian markets.

Stage	Focus
Initial launch	Greater Accra and selected high-demand service categories.
Expansion	Kumasi, Tema, Takoradi, Cape Coast, Tamale and other high-demand areas.
National scale	Broader Ghana coverage and expanded professional categories.
Regional scale	Potential expansion into other African markets after the Ghana model is proven.

Potential initial categories:

- Plumbing
- Electrical services
- Carpentry
- Masonry
- Painting
- Welding
- Cleaning
- Air-conditioning and appliance repair
- Auto services

5. Users, Actors and Roles

The platform should support role-based access while allowing a person to have multiple roles. For example, a professional could request a service and also offer services through the same account.

Actor	Primary responsibilities
Individual customer	Find services, create requests, compare providers, communicate, hire, pay and review.
Artisan / service provider	Create profile, define skills and service area, respond to requests, quote, perform jobs and receive payment.
Experienced professional	Offer consulting, advisory, mentoring, technical and project-based services.
Business / agency	Create organization account, manage team members, create projects and requests, source providers and manage engagements.
Solid Connect administrator	Manage users, verification, categories, requests, jobs, disputes, payments, content and analytics.

6. Core User Journeys

6.1 Customer

Open app

- > Register / Sign in
- > Select service
- > Describe requirement
- > Add location
- > Select timing
- > Add budget if applicable
- > Upload photos or video if needed
- > Submit request
- > Providers matched
- > Receive quotations
- > Compare providers
- > Select provider
- > Chat and coordinate
- > Job completed
- > Payment / confirmation
- > Review provider

6.2 Provider

Register

- > Create provider profile
- > Select skills and categories

- > Add experience and portfolio
- > Submit verification
- > Approval
- > Set service area and availability
- > Receive requests
- > Quote
- > Communicate
- > Complete job
- > Receive payment
- > Build rating and work history

6.3 Organization

Create organization

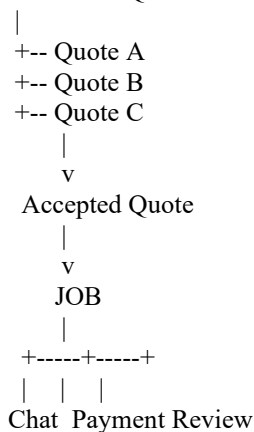
- > Add members and roles
- > Create project
- > Create workforce/service request
- > Receive applications / quotations
- > Select providers
- > Manage project
- > Approve work
- > Payment and reporting
- > Maintain provider relationship

7. Marketplace Mechanics

The recommended model is hybrid. Customers can search providers directly or submit a structured service request and allow Solid Connect to identify suitable providers.

Model	Description	Recommendation
Direct search	Customer searches and browses providers.	MVP
Service request	Customer describes a need and providers are matched.	Core MVP
Quotation / bidding	Providers respond with prices and proposals.	Core feature
Negotiation	Customer and provider agree details through controlled communication.	Core feature
Recurring service	Customer creates recurring arrangements.	Phase 2

SERVICE REQUEST



8. Provider Matching and GIS

Provider matching should be a core capability. Candidate providers can be ranked using service skill, distance, availability, verification, rating, experience, response history and price suitability.

Factor	Illustrative weight
Skill match	30%
Distance	20%
Availability	15%
Rating	10%
Experience	10%

Verification	5%
Price suitability	5%
Response history	5%

The weights above are illustrative and should be validated using real marketplace data.

Customer GPS

- > NestJS API
- > PostGIS spatial query
- > Providers within service radius
- > Filter by skill, availability and verification
- > Rank candidates
- > Return matches

PostGIS can support provider discovery, distance calculations, service areas, demand analysis and future marketplace intelligence.

9. Trust, Verification and Reputation

Trust should be one of Solid Connect's core pillars. Provider profiles should communicate identity, qualifications, experience, portfolio, completed jobs and reputation.

Level	Purpose
Registered	Provider has created an account.
Identity verified	Identity information has been reviewed.
Profession verified	Relevant qualifications or professional information have been verified where applicable.
Experience verified	Employment, references or work history have been reviewed where applicable.
Solid Connect verified / certified	Higher-level verification or assessment under a defined Solid Connect standard.

- Overall rating
- Quality of work
- Professionalism
- Communication
- Punctuality
- Value for money
- Completed jobs
- Cancellation and response history

10. Payments and Commercial Transactions

Payment architecture should be modular so that Solid Connect can support multiple payment methods and providers. For Ghana, mobile money and card payments are important considerations.

Customer

- > Payment gateway
- > Solid Connect transaction
 - > Platform commission
 - > Provider payout
- > Refund / dispute process where applicable

An escrow-style model may be considered for suitable jobs, subject to the selected payment provider, business rules and legal requirements.

11. Disputes, Safety and Marketplace Protection

Complaint

- > Evidence submitted
- > Solid Connect review
- > Decision
 - + Provider paid
 - + Partial refund
 - + Full refund
 - + Warning / suspension
 - + Further investigation

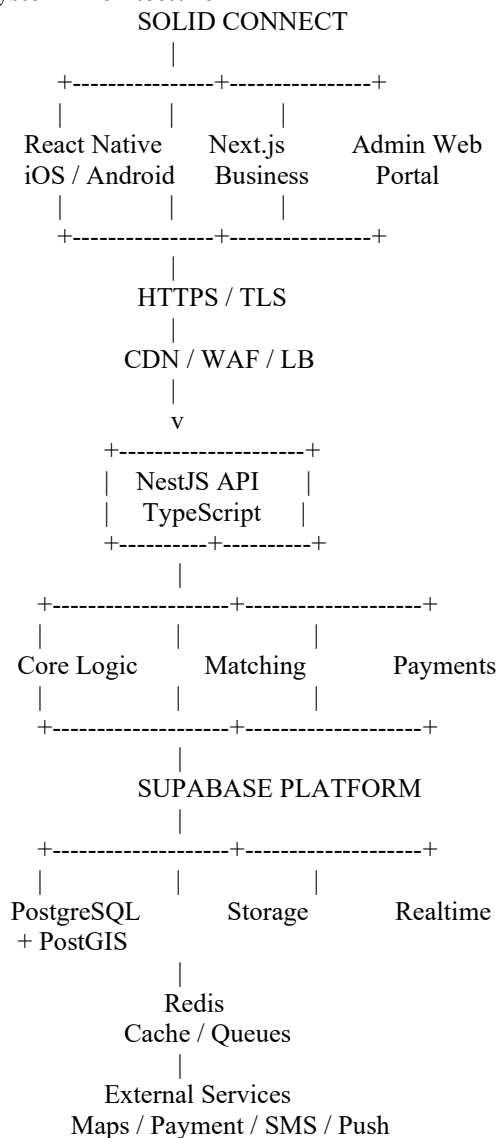
- Allow users to report providers and customers.

- Maintain auditable job and payment events.
- Base verified reviews on completed platform jobs.
- Introduce controls against fake accounts, spam and payment abuse.

12. Core Product Features

Module	Core features
Authentication	Phone/email registration, OTP, login, sessions and recovery.
Profiles	Customer, provider and organization profiles.
Provider marketplace	Skills, categories, experience, portfolio, service areas, availability, reviews and verification.
Service requests	Description, location, timing, budget and media attachments.
Matching	Location and skill matching with ranking.
Quotes	Provider quotation, comparison and acceptance.
Jobs	Job lifecycle and completion confirmation.
Chat	Customer-provider and organization-provider communication.
Payments	Payment status, commissions, payouts and refunds.
Reviews	Post-job ratings and reviews.
Notifications	Push and in-app notifications.
Organizations	Business profiles, members, projects and workforce requests.
Administration	Users, verification, disputes, payments, categories, reports and analytics.

13. System Architecture



14. Recommended Technology Stack

Layer	Technology	Purpose
Mobile	React Native + TypeScript + Expo	Cross-platform iOS and Android app.
Navigation	React Navigation	Screen navigation and role-based flows.
Mobile state	Zustand	Lightweight local state.
Server state	TanStack Query	Caching, fetching and synchronization.
Business web	Next.js + React + TypeScript	Business and agency portal.
Admin web	Next.js + React + TypeScript	Administration and operations.
Backend	NestJS + Node.js + TypeScript	Core business logic and REST API.
Database	Supabase PostgreSQL	Transactional application data.
GIS	PostGIS	Location and spatial queries.
Authentication	Supabase Auth	Identity and sessions.
Storage	Supabase Storage	Images, documents and portfolio media.
Realtime	Supabase Realtime	Chat and selected live updates.
Cache / queue	Redis + BullMQ	Caching, rate limiting and background jobs.
Search	PostgreSQL initially, OpenSearch later	Provider and service search.
Maps	Google Maps or Mapbox	Maps, geocoding and location.
Notifications	FCM + APNs	Android and iOS push.
Infrastructure	Supabase + AWS/Cloudflare as required	Scalable cloud infrastructure.
CI/CD	GitHub Actions	Automated testing and deployment.
Monitoring	Sentry + OpenTelemetry	Application observability.

15. Supabase Strategy

Supabase is recommended as a managed platform foundation rather than the location for all application logic. Its PostgreSQL base and PostGIS support are especially useful for a location-driven marketplace.

Capability	Recommended use
PostgreSQL	Core transactional data.
PostGIS	Geographic matching and spatial analysis.
Auth	Authentication and sessions.
Storage	Profile photos, portfolios and documents.
Realtime	Chat and selected live updates.
Edge Functions	Lightweight server-side functions where appropriate.

Key architecture principle

The mobile applications should not contain critical marketplace business rules. React Native should communicate with the NestJS backend, while NestJS manages business logic and accesses Supabase securely.

16. Architecture for Approximately 1 Million Users

The one-million-user target should be treated as a capacity objective. It does not require deploying a large microservices environment from day one. The system should scale horizontally as actual traffic grows.

Internet

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CDN / WAF

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Load Balancer

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+-- API 1

+-- API 2

+-- API N

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Core Services

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+-- PostgreSQL + PostGIS

+-- Redis

+-- Search

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Read replicas / backups as required

- Horizontal API scaling.

- Connection pooling and efficient indexing.
- Read replicas when database reads require them.
- Background queues for notifications and non-critical work.
- Redis caching and rate limiting.
- Object storage and CDN for media.
- OpenSearch when search volume or complexity justifies it.
- Monitoring and optimization based on real usage.

Scale stage	Architecture direction
0 to 100,000 users	Supabase foundation, NestJS modular backend, React Native, basic Redis, CDN and monitoring.
100,000 to 1,000,000 users	Multiple API instances, workers, stronger caching, database read capacity, search and enhanced monitoring.
1 to 10 million users	Evaluate specialized services, event streaming, dedicated search, data warehouse, Kubernetes or multi-region deployment based on actual demand.

17. Backend and Microservices Strategy

A modular monolith is recommended initially. NestJS can separate the backend into clear modules while keeping deployment and development manageable.

solid-connect-api

```
+-- auth
+-- users
+-- roles
+-- providers
+-- services
+-- categories
+-- requests
+-- matching
+-- quotes
+-- jobs
+-- contracts
+-- payments
+-- reviews
+-- verification
+-- organizations
+-- projects
+-- chat
+-- notifications
+-- disputes
+-- admin
```

Individual modules can later be extracted into independent services if a component such as search, chat, notifications, matching or payments develops an independent scaling requirement.

18. Security Architecture

- TLS for data in transit.
- Secure authentication and token management.
- Role-based access control.
- Multi-factor authentication for sensitive administrative accounts.
- API rate limiting and abuse protection.
- Input validation and secure file uploads.
- Encrypted storage and secure handling of sensitive documents.
- Audit logs for administrative, payment and verification events.
- Dependency and vulnerability scanning.
- Automated backups and tested recovery procedures.

19. Administration and Operations

Area	Capabilities
Dashboard	Users, providers, organizations, active jobs, revenue, verification and disputes.
Users	Search, view, edit, suspend and manage accounts.
Providers	Verification, skills, portfolios, service areas and performance.
Requests and jobs	Monitor lifecycle and investigate issues.
Payments	Transactions, commissions, payouts, refunds and exceptions.
Verification	Review and approve or reject documents.
Disputes	Review complaints, evidence and outcomes.
Content	Categories, service types and platform settings.
Analytics	Marketplace, demand, supply and geographic trends.

20. Analytics and Marketplace Intelligence

KPI	Purpose
Requests created	Measures demand.
Requests fulfilled	Measures marketplace effectiveness.
Provider response rate	Measures supply engagement.
Time to first response	Measures responsiveness.
Time to hire	Measures efficiency.
Job completion rate	Measures reliability.
Cancellation rate	Measures operational health.
Average transaction value	Supports pricing and revenue analysis.
Platform commission	Measures monetization.
Customer retention	Measures repeat usage.
Provider retention	Measures supply health.

21. Future AI Opportunities

- Natural-language service classification.
- Intelligent provider matching.
- Price estimation.
- Fraud and abuse detection.
- Customer support assistant.
- Demand and provider-supply forecasting.

AI should be introduced after the core marketplace is validated and sufficient quality data is available.

22. Recommended MVP Scope

Feature	MVP	Phase 2	Phase 3
Registration and authentication	Yes		
Customer profiles	Yes		
Provider profiles	Yes		
Provider verification	Yes		
Service categories	Yes		
Service requests	Yes		
Location and GIS matching	Yes		
Quotes	Yes		
Chat	Yes		
Jobs and completion	Yes		
Reviews	Yes		
Payments	Yes, subject to final design		
Business / agency accounts		Yes	
Projects and workforce requests		Yes	
Recurring services		Yes	
Advanced analytics		Yes	
OpenSearch		As required	
AI matching			Yes

Predictive pricing			Yes
Advanced workforce optimization			Yes

23. API and Integration Architecture

The API should use versioned REST endpoints with strong validation, authentication, authorization and OpenAPI documentation.

/api/v1/auth
 /api/v1/users
 /api/v1/providers
 /api/v1/services
 /api/v1/categories
 /api/v1/requests
 /api/v1/quotes
 /api/v1/jobs
 /api/v1/payments
 /api/v1/reviews
 /api/v1/chat
 /api/v1/notifications
 /api/v1/organizations
 /api/v1/projects
 /api/v1/admin

24. DevOps, Deployment and Reliability

Developer

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GitHub

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Pull Request

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Automated tests and security checks

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Build

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Staging

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Approval

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Production

- GitHub Actions for CI/CD.
- Docker for reproducible builds.
- Separate development, staging and production environments.
- Terraform for cloud infrastructure where appropriate.
- Automated database backups and recovery testing.
- Application monitoring, logs and error tracking from the beginning.

25. Legal, Compliance and Governance

The exact legal and regulatory requirements should be confirmed with qualified legal and compliance professionals before launch.

- Terms of service and user agreements.
- Privacy policy and data protection requirements.
- Provider terms and service arrangements.
- Organization and project contracting arrangements.
- Payment, refund and dispute policies.
- Identity and professional verification policies.
- Data retention and deletion procedures.
- Professional licensing requirements for regulated services.
- Consumer protection and marketplace responsibilities.

26. Key Product and Technical Risks

Risk	Mitigation
Low marketplace liquidity	Launch in a focused geography and limited number of high-demand categories.
Unverified providers	Use staged identity, qualification and experience verification.
Poor service quality	Use completed-job reviews, performance history and dispute management.
Off-platform transactions	Provide useful in-app communication, payment and job management.
Payment disputes	Define clear payment, completion and refund rules.
Database bottlenecks	Use indexing, pooling, caching, read capacity and background processing.
Media growth	Use object storage and CDN.
Over-engineering	Start modular and scale components based on actual usage.
Security threats	Use WAF, RBAC, MFA for admins, rate limiting, audit logs and security testing.

27. Proposed Development Roadmap

Stage	Focus
1	Requirements validation, business rules, actors and user journeys.
2	UI/UX design, information architecture and prototype.
3	Database ERD, API specification and technical foundation.
4	Authentication, profiles, categories and provider onboarding.
5	Requests, matching, quotations, jobs and chat.
6	Payments, reviews, verification and notifications.
7	Business portal and administration portal.
8	Testing, security review, performance testing and production preparation.
9	App Store and Google Play deployment.
10	Post-launch monitoring, optimization and feature expansion.

28. Recommended Next Steps with Solid Connect Limited

Before development begins, the following decisions should be finalized:

1. Confirm the primary launch market and initial service categories.
2. Confirm the exact user types and whether users can have multiple roles.
3. Confirm whether the initial product is an artisan marketplace, professional services marketplace, workforce marketplace, or phased combination.
4. Confirm provider verification and who performs verification.
5. Confirm payment processing, commission rules and whether escrow is required.
6. Confirm the organization and agency workflow.
7. Confirm MVP features and launch priorities.
8. Confirm branding, domain, app ownership and Apple App Store and Google Play accounts.
9. Confirm data ownership, hosting preferences and operational responsibilities.
10. Confirm legal, privacy and compliance requirements.
11. Approve the technical architecture and proceed to detailed requirements and database design.

29. Recommended Architecture Summary

Recommended technology direction

React Native + Expo for mobile, Next.js for business and admin web, NestJS + TypeScript for the core backend, Supabase PostgreSQL + PostGIS for the data foundation, Supabase Auth/Storage/Realtime for selected managed services, Redis for caching and background jobs, and AWS/Cloudflare infrastructure as scale and operational requirements increase.

This approach balances development speed, maintainability, geographic capability, security and future scalability. It avoids unnecessary microservices complexity at launch while preserving a clear path to specialized services and larger infrastructure as Solid Connect grows.

Appendix A: Conceptual Data Entities

Entity	Purpose
User	Core identity record.
Role	Customer, provider, professional, organization and administrator capabilities.
Profile	Public or private user information.
Provider Profile	Skills, experience, service areas, availability and reputation.
Skill	Specific capability offered by a provider.

Service Category	Grouping of services.
Service Request	Customer or organization request.
Quote	Provider response to a request.
Job	Accepted service engagement.
Contract	Longer-term or formal engagement.
Organization	Business or agency account.
Project	Organization-level work grouping.
Payment	Payment and payout record.
Review	Customer or organization feedback.
Message	Chat communication.
Notification	System notification.
Verification	Identity, qualification or experience verification.
Document	Uploaded verification or project document.
Dispute	Formal marketplace complaint.

Appendix B: Conceptual Architecture Flow

CUSTOMER / PROVIDER / ORGANIZATION

